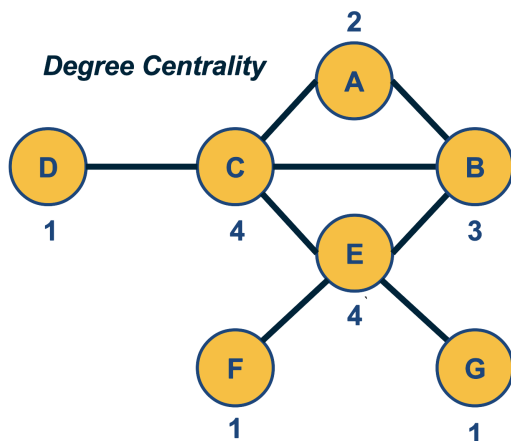


L6: Degree, Eigenvector and The Katz Centrality

Degree and Strength Centrality



The simplest way to define the importance of a network node is based on its number of connections – the more connections a node has, the more important it is. So, the **degree** centrality of a node is simply the degree of that node.

For weighted networks, the corresponding metric is the sum of the weights of all edges of that node, i.e., the **strength** of that node.

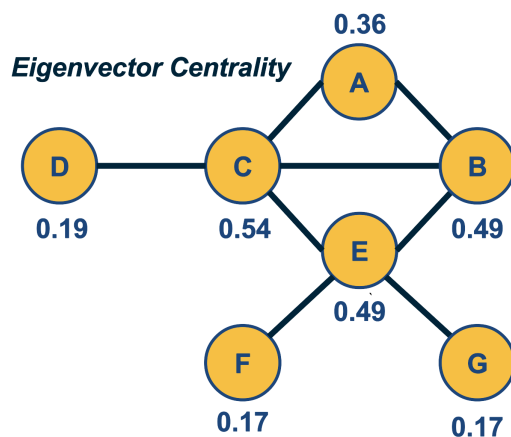
For directed networks, we can separate between in-degree and out-degree centrality.

One problem with this definition of centrality however is that it only captures the “local role” of a node in the network. A node may have many connections within an isolated cluster of nodes that are completely disconnected from the

rest of the network. On the other hand, an important node may have only a few direct connections but it can be the only node between two large groups of nodes that are otherwise disconnected.

Another issue with this centrality metric is that it is easy to manipulate locally. Back in the early days of the Web, some search engines used to rank results based on degree centrality (*how many other Web pages point to that page*). Some online firms quickly exploited that vulnerability: they would create 1000s of links from fake web pages so that they boost the centrality/rank of their sponsored web sites.

Eigenvector Centrality



A better centrality metric is to consider not only the number of neighbors of a node -- but also the centrality of those neighbors. So the basic idea is that a node is more central when its neighbors are also more central. Suppose that we are given an undirected network with adjacency matrix A . Let v_i be the centrality of node i . We can then define that:

$$v_i = \frac{1}{\lambda} \sum_j A_{i,j} v_j$$

We can think of the term $\frac{1}{\lambda}$ as a normalization constant for now. Note that the centrality of a node is the (normalized) sum of the centralities of all its neighbors.

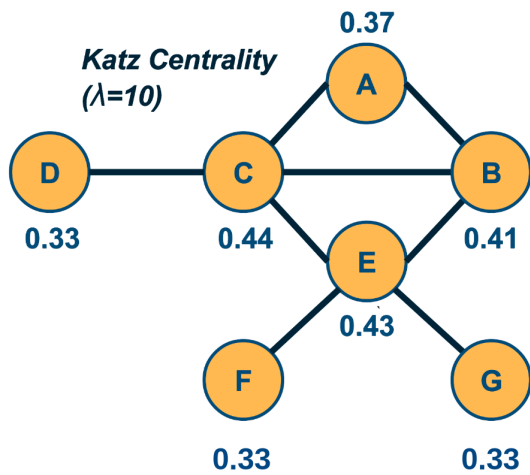
The previous equation can be written in matrix form as:

$$\lambda v = A v$$

where v is the vector all node centralities. This equation, however, is simply the **eigenvector** definition of matrix A : λ is the corresponding eigenvalue for eigenvector v . This is why we refer to this centrality metric as “**eigenvector centrality**”.

We wish to have non-negative centralities. In other words, the eigenvector v corresponding to eigenvalue λ should consist of non-negative entries. It can be shown (using the Perron–Frobenius theorem) that using the largest eigenvalue of A satisfies this requirement.

The Katz Centrality



The Katz centrality metric is a variation of eigenvector centrality that is more appropriate for directed networks.

The eigenvector centrality may be zero even for nodes with non-zero in-degree and out-degree, and so Katz starts from the same equation as eigenvector centrality but it also assigns a small centrality β to every node.

So, the definition becomes:

$$v_i = \beta + \frac{1}{\lambda} \sum_j A_{ij} v_j$$

where the summation is over all nodes j that connect with i .

Given that we are only interested in the relative magnitude of the centrality values, we can arbitrarily assign $\beta=1$.

Then, we can rewrite the previous definition in matrix form:

$$\mathbf{v} = (\mathbf{I} - \frac{1}{\lambda} \mathbf{A})^{-1} \cdot \mathbf{1}$$

Where $\mathbf{1}$ is a n -by-1 vector of all ones.

The value of λ controls the relative magnitude between the constant centrality β we assign to each node and the centrality that each node derives from its neighbors. If λ is very large, then the former term dominates and all nodes have roughly the same centrality. If λ is too small, on the other hand, the Katz centralities may diverge. This is the case when the determinant of the matrix $(\mathbf{I} - \frac{1}{\lambda} \mathbf{A})$ is zero, which happens when λ is equal to an eigenvalue of $\mathbf{A}$. To avoid this divergence, the value of λ is typically constrained to be larger than the maximum eigenvalue of $\mathbf{A}$.

Note that the Katz centrality values given in the figure are normalized by the L2-norm of the centrality vector.

Food For Thought

Derive the matrix-form equation of Katz centrality from the initial definition.